



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

PERSONAL EXAM MARK TRACKER

CAPSTONE PROJECT PRESENTATION
PROGRAMMING IN JAVA - CSA0901

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INTRODUCTION

PERSONAL EXAM MARKS TRACKER IS A JAVA-BASED APPLICATION DEVELOPED TO HELP STUDENTS MONITOR AND ANALYZE THEIR ACADEMIC PERFORMANCE ACROSS MULTIPLE EXAMINATIONS.

THE SYSTEM ALLOWS USERS TO REGISTER, LOG IN, ENTER EXAMINATION MARKS, AND VIEW DETAILED PERFORMANCE REPORTS.

BY COMPARING MARKS FROM DIFFERENT EXAMS, IT IDENTIFIES IMPROVEMENTS, DECLINES, AND LEARNING TRENDS, WHILE ALSO PROVIDING STUDY RECOMMENDATIONS, TARGET PREDICTIONS, ACHIEVEMENT BADGES, AND OVERALL PERFORMANCE ANALYSIS.

THE PROJECT AIMS TO HELP STUDENTS UNDERSTAND THEIR STRENGTHS AND WEAKNESSES AND SUPPORT CONTINUOUS ACADEMIC IMPROVEMENT.

LITERATURE REVIEW

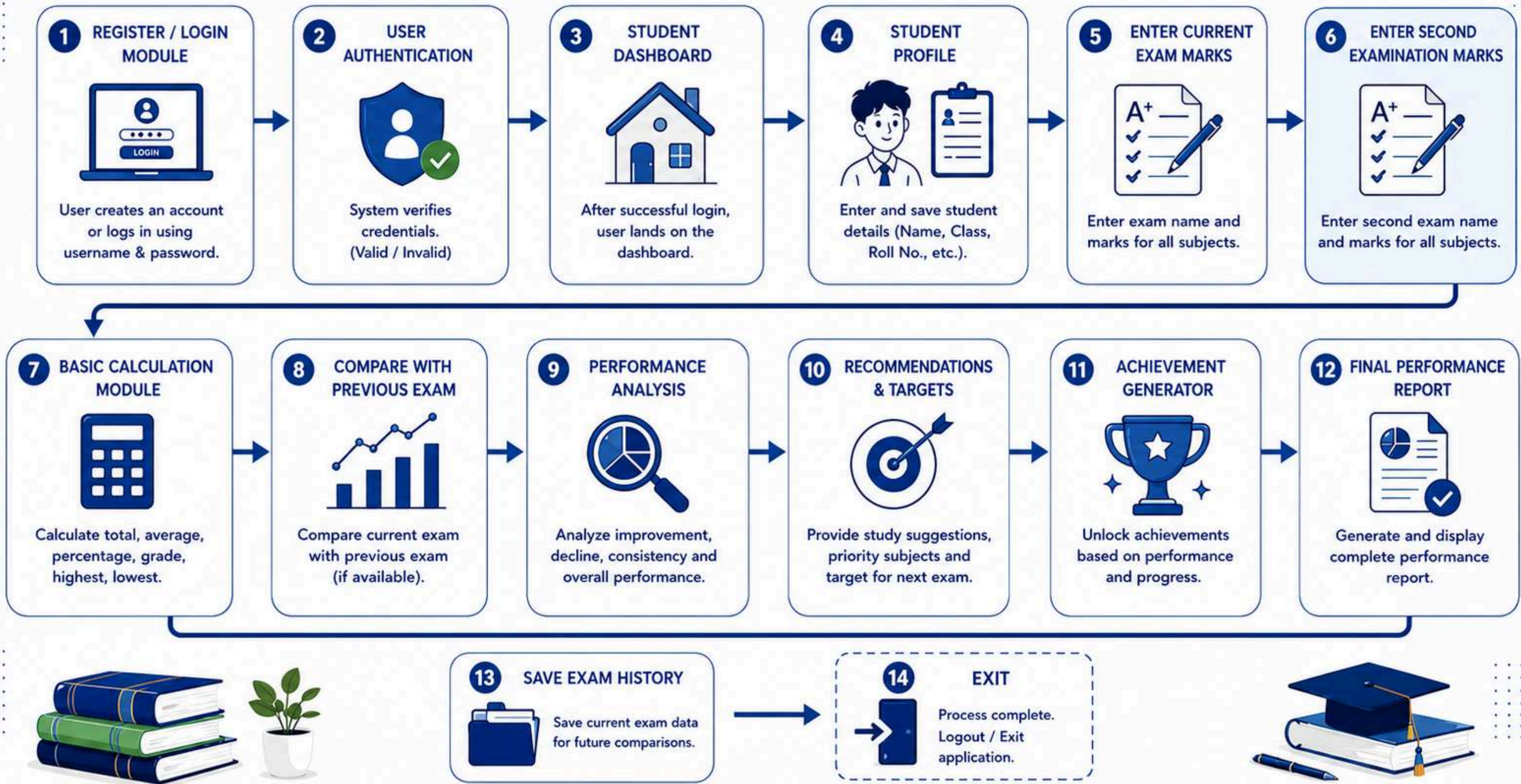
S.N O	RESCREACH PAPER	AUTHOR	ADVANTAGES	DISADVANTAGES	TECHNOLOGIES USED
1	Analyzing Student's Academic Performance Using Educational Data Mining	M.S. Islam et al.	Identifies "at-risk" subjects early by tracking historical trend data.	Relies heavily on consecutive internal exam consistency.	Educational Data Mining (EDM), Classification Classifiers
2	Peer-Assessed Automated Grading System	J. Shi & L. Zhang	Cuts down manual workload through secure, cloud-based batch processing.	Struggles with highly creative or nuanced qualitative metrics.	ASP.NET MVC, Chart.js, Cloud Architecture
3	Learning Analytics Dashboard for Motivation and Performance	M. Matchin et al.	Boosts student drive through goal tracking and smart peer comparisons.	Peer comparison can sometimes induce anxiety in lower-performing brackets.	Learning Analytics (LA), Grade Predictive Modelling
4	Automated Grading System with Student Performance Analytics	B.E. Bato & J.R. Pomperada	- Instantly calculates grades and removes human calculation errors. - Feeds live scores directly into predictive data dashboards.	- Relies heavily on educators adapting to rigid online grading rubrics. - Demands secure, encrypted server-side rollback backups.	Laravel Web Framework, Role-Based Access Control (RBAC), Data Encryption Engines



METHODOLOGY



- Requirement Analysis: Identify tracking parameters (subjects, term marks, total scale).
- Data Structuring: Create ExamRecord and Subject objects stored using Java ArrayList.
- User Interaction: Interactive console/GUI prompts accepting student marks via Scanner.
- Analytics & Evaluation: Compute aggregates, pass/fail status, and grade classifications.
- Report Generation: Summarize subject trends, cumulative percentages, and averages.
- Testing: Validate boundary inputs (e.g., negative marks or values exceeding 100).



USER AUTHENTICATION & PROFILE MANAGEMENT



MODULE OVERVIEW

This module handles the initial user access to the system and manages basic student credentials.



ROLE

Manages initial user access and stores basic student credentials.

1 USER REGISTRATION



User registers by providing basic details.

- Name
- Grade
- School
- Username
- Password



2 ACCOUNT VALIDATION & LOGIN CHECK



System validates credentials and verifies user identity.

- Username & Password Verification
- Secure Login Check



3 PROFILE OVERVIEW DASHBOARD



After successful login, user is directed to the profile dashboard.

- View Profile Summary
- Quick Access to System Modules
- Recent Activity Overview

EXAM & SUBJECT DATA ENTRY



MODULE OVERVIEW

This module captures term-wise subject names and numerical marks for each exam.



ROLE

Captures term-wise subject names and numerical marks.

1

DATA STRUCTURE INITIALIZATION



Initialize dynamic arrays to store subjects and marks.

- `String` subjects[]
- `int` exam1Marks[]
- `int` exam2Marks[]

Example:

```
subjects[] = {"Maths", "Science", "English", "Social Science"}
exam1Marks[] = { 0, 0, 0, 0 }
exam2Marks[] = { 0, 0, 0, 0 }
```

2

SCORE INPUT VALIDATION



Validate each score to ensure it is a number within the allowed range.

- Accept only numerical input
- Ensure marks are within 0 to 100
- Display error for invalid entries
- Re-prompt until valid input is provided

3

EXAM-WISE RECORD GENERATION



Capture and store marks separately for each exam.

- Enter marks for all subjects in Exam 1
- Enter marks for all subjects in Exam 2
- Store data for comparison and analysis in next module

MODULE 3

CORE EVALUATION & AGGREGATE ANALYTICS

MODULE OVERVIEW

This module computes key statistical metrics and determines overall performance across examination terms.

ROLE

Computes fundamental statistical metrics across examination terms.

1 TOTAL SCORE & AVERAGE CALCULATION



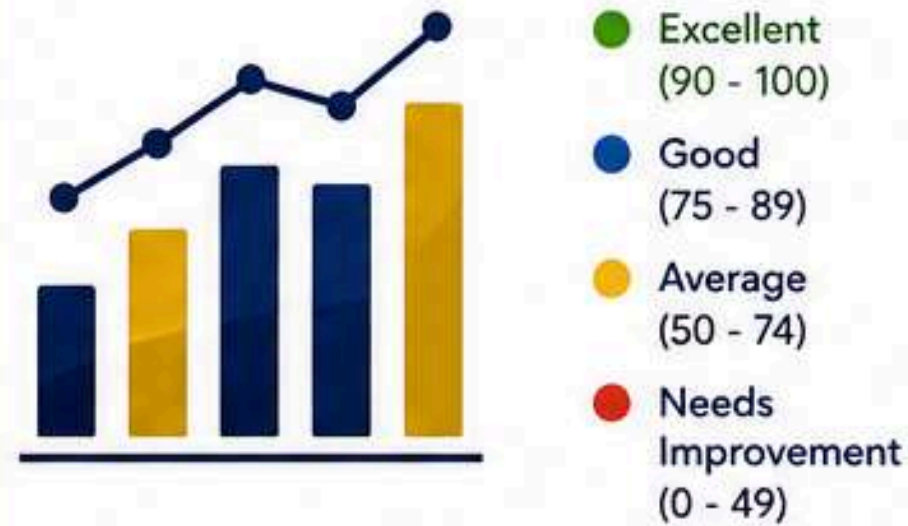
Calculate total score and average percentage for each exam.

- Sum of marks of all subjects (Exam 1 & Exam 2)
- Maximum possible score = Number of subjects × 100
- Average Percentage = $\frac{\text{Total Obtained}}{\text{Maximum Score}} \times 100$

Example

Total Obtained : 812 / 10 subjects
Maximum Score 1000
Average Percentage : **81.20%**

2 SUBJECT-WISE PERFORMANCE CATEGORIZATION



Categorize performance for each subject in both exams.

- Compare marks of each subject against defined ranges
- Assign performance category (Exam 1 & Exam 2)
- Helps in identifying strengths and weak areas

Example (Exam 1)

Subject	Marks	Category
Maths	92	Excellent
Science	78	Good
English	63	Average
Social Science	45	Needs Improvement

3 OVERALL PERFORMANCE GRADE & STATUS



Determine overall grade and status based on average percentage.

- Grade is assigned using predefined grading scale
- Status provides a quick summary of performance level
- Encourages continuous improvement

Average %	Grade	Status
90 - 100	A+	Outstanding
75 - 89	A / B	Very Good
50 - 74	C	Satisfactory
0 - 49	D	Needs Improvement

PERFORMANCE COMPARISON & TREND ANALYSIS



MODULE OVERVIEW

This module analyzes progress and tracks performance shifts between two sequential exams.

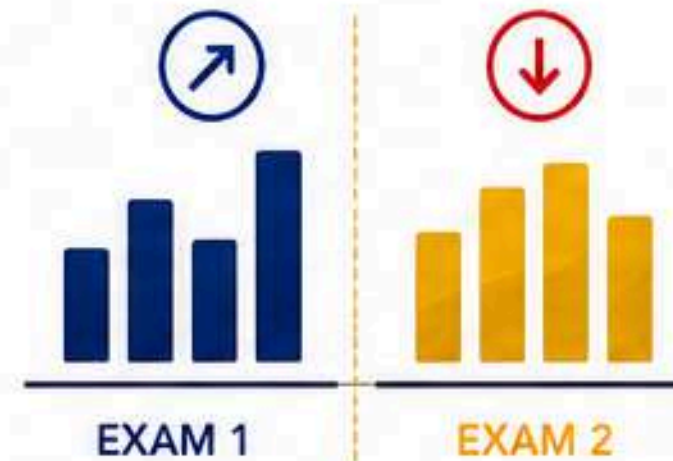


ROLE

Analyzes progress and tracking shifts between sequential exams.

1

IMPROVEMENT & DROP DETECTOR



Identifies the highest score gain and the sharpest decline.

↑ **biggestImprovement**
(Highest positive difference)

↓ **biggestDrop**
(Most negative difference)

Example

Biggest Improvement : Maths (+18)

Biggest Drop : Social Science (-12)

2

CONSISTENCY ANALYZER

SUBJECT	EXAM 1	EXAM 2	STATUS
Maths	72	90	↑ Improved
Science	65	63	↓ Declined
English	78	78	— Unchanged
Social Science	58	70	↑ Improved
Computer Science	85	80	↓ Declined

Compares subject-wise marks and classifies outcomes.

↑ **Improved** : Exam 2 > Exam 1

↓ **Declined** : Exam 2 < Exam 1

— **Unchanged** : Exam 2 = Exam 1

3

IMPROVEMENT SCORE CALCULATION



IMPROVEMENT SCORE

+28.57%

Generates an overall percentage score indicating positive performance shift.

Calculation Logic

$$\text{Improvement Score (\%)} = \frac{(\text{Sum of Positive Differences})}{(\text{Sum of Absolute Differences})} \times 100$$

- Positive value indicates overall improvement
- Negative value indicates overall decline

MODULE 5

SMART ANALYTICS, GAP ANALYSIS & TARGET PREDICTION



MODULE OVERVIEW

This module delivers automated insights, study priorities, and future score targets to help students improve strategically.



ROLE

Delivers automated insights, study priorities, and future score targets.

1

MARKS GAP ANALYSIS

Calculates distance from target score limits per subject.

SUBJECT	Exam-2	Exam-1	GAP	STATUS
Maths	72	90	18	↓
Science	63	85	22	↓
English	78	80	2	—
Social Science	70	80	10	↓
Computer Science	80	90	10	↓

- On Track (Gap ≤ 5)
- Needs Work (5 < Gap ≤ 15)
- Needs Attention (Gap > 15)



Helps students understand how close they are to their target in each subject.

2

SMART STUDY RECOMMENDATION & PRIORITY GENERATOR

Ranks subjects that need immediate academic focus.



PRIORITY LIST

- 1 Science (Gap: 22)
- 2 Maths (Gap: 18)
- 3 Computer Science (Gap: 10)
- 4 Social Science (Gap: 10)
- 5 English (Gap: 2)



Analyzes gaps, performance trends and consistency.



Ranks subjects by priority level (High / Medium / Low).



Suggests focus areas for better preparation.

3

TARGET PREDICTION & ACHIEVEMENT SYSTEM

Predicts potential future scores and rewards achievements.



TARGET PREDICTION

Based on current trend, your predicted score in next exam:

87.4%

ACHIEVEMENT SYSTEM



Consistent Performer



Improvement Achiever



High Scorer



Goal Getter

- ✓ Predicts future scores using improvement trend.
- ✓ Unlocks badges based on overall performance.
- ✓ Motivates students to achieve their targets.

CHALLENGES & LIMITATIONS

- Only compares available exam data — comparison and trend analysis require marks from at least two exams.
- No database storage — the current version does not permanently store exam records after the program ends.
- Fixed grading system — grades are based on predefined percentage ranges and cannot be customized.
- Single-user focused — the application is designed for one student's performance tracking.
- No automatic exam data import — marks cannot be directly imported from a school/college system or spreadsheet.
- No GUI: The system lacks a graphical interface, making it less user-friendly and harder to use compared to a web-based application.

FUTURE ENHANCEMENTS

- Graphical User Interface (GUI) – Develop a user-friendly frontend instead of a console-based interface.
- Database Integration – Store student profiles and exam records permanently.
- Multiple Exam Tracking – Allow students to add and compare any number of examinations.
- Performance Charts – Display marks and progress using graphs and charts.
- Advanced AI Prediction – Use AI/ML to provide more accurate score predictions and personalized study recommendations.
- Detailed Reports – Generate downloadable performance reports for students and parents.

CONCLUSION

The project will provide students with more than just their marks. It transforms raw scores into meaningful insights, helping them understand their academic progress, recognize strengths and weaknesses, and make informed decisions about what to improve before the next exam.

Rather than functioning as a simple marks calculator, the Personal Exam Performance Analyzer serves as a student-centric academic performance tracking and decision-support tool that encourages continuous improvement and smarter study planning

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THANK YOU